

The Forge — game design v0.2 (mechanical detail)

A real-world, festival-scale strategy game. This version is built on the honest real-world constraints (see `REAL-WORLD.md`): players are civilians, not gamers; custom IoT at scale is fragile; the field has no infra; safety/crowd-flow lead. So the structure is a **megagame**, not a swarm of smart totems.

What changed from v0.1 (the improvements):

1. **Command layer added.** Strategy is run by ~8 **captains** per faction, not by 150 phones. Foot soldiers follow squad leaders. The complex layer (economy/intel/parley) lives at HQ.
2. **Capture is presence-by-pulse with caps**, not vague "tapping." Distinct bodies, stacking-capped, with decay — which kills blobbing, turtling, and one-runner-cheese in one move.
3. **Strongholds are invulnerable until the siege phase** → early rush is impossible (the sim showed rush dominance; this is the structural fix).
4. **Battering** makes the siege endgame reachable (the "third lever" the sim found missing): you need bodies *and* a supply dump.
5. **Sensing is decoupled from rules.** A capture event is the same whether it comes from an NFC tap, a marshal's headcount, or a beacon. The design never bets on one fragile tech.
6. Foot-soldier engagement via **physical carry** (supply / intel / bridge-kit tokens) → escort & interception gameplay, not just walking.

Baseline numbers (one real run)

- ~150 players / faction, 2–4 factions · 12 sectors + 2–4 strongholds ·
- Resource unit = **Supply**. Pilot scales this down (60 players, 8 sectors).

90-minute session · economy resolved per minute, control per 10 s.



1. Control & capture (the core mechanic, with numbers)

Each sector has a **control meter** from **−100 (blue)** to **+100 (red)**.

- A **presence pulse** = a *distinct* faction badge registered at the station

(tap, or marshal headcount). A pulse counts as "present" for **60 s** (re-tap to stay). Rate-limited: 1 pulse / badge / station / 20 s (anti-spoof).

- **Net presence** $P = (\text{red present} - \text{blue present})$, weighted by class, ****capped** at $\pm 5^{**}$ (a 6th body adds nothing → forces spreading out, kills blobbing).
- **Meter step (per 10 s):** $\Delta C = P \times \text{capRate}$, $\text{capRate} = 6$.

- a 3-net edge flips a neutral sector in ~55 s. Feels like a real push.
- **Flip** ownership at ±60 (hysteresis → no flicker around 0).
 - **Decay**: uncontested sectors drift toward 0 at 3 / 10 s → you can't hold

2. Command layer (how 150 people are playable)

- Faction = ~8 **captains (HQ)** + foot soldiers in **squads** of ~10 under a **squad leader**.
- HQ issues **orders** ("Squad 3 → take sector D"); the squad leader's app shows

3. Economy (Supply)

Source	Rate
held sector	+10 / min
stronghold	+30 / min
Quartermaster on a sector	+50% that sector
per-sector yield cap	flat regardless of how many stack (anti-snowball)
Sink	Cost
Build bridge	120 (~2–3 min income)

4. Classes (composition matters)

Class	Presence weight	Move	Ability (cost / cooldown)
Runner	0.5	fast	Scout : reveal a region to HQ (free, 60 s cd); carries intel chits 2×
Anchor	2.0 def / 1.0 atk	slow	Dig In : +50% defensive ΔC on this sector 90 s (15 Supply, 4 min cd)
Engineer	1.0	normal	Build/Sabotage bridge (kit + 120 Supply; sabotage

5. Intel warfare (physical + concrete)

the whole map passively; holding everything means spreading thin = exposed. (Anti-snowball + anti-turtle, built into one rule.)

This is exactly the net-pressure model already in the sim — these are its levers.

- the order + a compass arrow; foot soldiers just follow.
- **Order latency** ≈ 30–60 s (comms cost). You set *intent*, you don't micromanage. The mesh/app latency becomes a **designed constraint**, not a bug.

Sink	Cost
Recon report (reveal a region 2 min)	20
Comms jam (black out enemy region 3 min)	80
Battering (siege, phase 4)	10 Supply → −1 enemy Wall
Rally / Dig-In ability fuel	15

Decision: **tempo** (spend to expand) vs **bank** (save for siege) vs **comeback**.

Class	Presence weight	Move	Ability (cost / cooldown)
			closes one 3 min, 1×
Quartermaster	1.0	normal	+50% yield on their sector; carries 2× supply tokens
Herald	1.0	normal	Parley (open a deal channel); Rally : +1 ally presence here 60 s (15, 5 min cd)

All-Anchor can't take ground (atk 1.0, slow); all-Runner can't hold (0.5). Forces a mix.

- HQ map is **fogged**: a sector's control is known only if you're present **or** scouted it in the last **2 min**.
- **Scout** (Runner) / **Recon report** (20 Supply) reveal.
- **Intel chits**: a held sector drops a physical "sigint" token every **3 min**;

6. Bridges

Fixed crossing gates, closed by default. **Build** = Engineer present + bridge-kit prop (carried from HQ) +

7. Siege & endgame (reachable, paced)

- Strongholds have **Wall = 100** and are **uncapturable while Wall > 0** AND only batterable in **Phase 4** (so no early rush — the structural fix).
- **Delay tower**: before you may batter a stronghold you must ****hold** a nearby

8. Parley & diplomacy

Heralds open a **deal channel** between HQs: truce (no contest on a named sector for N min), trade

9. Scoring

Per minute: **+1 / held sector**, **+3 / stronghold**. Objectives: GM supply-drop **+15**, intel chit cashed **+2**,

10. The 90-minute arc (concrete triggers)

Time	Phase	Unlocks / events
0–15	Muster & landgrab	strongholds safe; neutrals open; income ramps
15–45	Contact	bridges buildable; intel war; GM drops a supply crate midfield
45–70	Escalation	comms-jam + parley unlock; GM drops a high-value objective; leader

- a Runner carries it to HQ → cash for a reveal or +5 Supply → **escort & interception** gameplay for foot soldiers.
- **Comms jam** (80): an enemy region goes dark on their map 3 min (captains blind).
 - **Deception**: don't tap when passing → stay hidden; Herald can plant a ****feint flag**** that shows false activity to scouts.

120 Supply → marshal opens it; **shared** (anyone crosses). **Sabotage** closes it 3 min. Opens a flank → shortens the path to enemy ground → momentum swing.

- delay tower for 3 continuous minutes****** ("lower the drawbridge"). Paces the endgame, stops a chaotic instant rush.
- **Battering**: with net presence at the stronghold, pour Supply: ****10** → **−1**
- Wall. **At Wall 0 it's capturable by control. Taking it = +50******** + the region's income; holding ALL enemy strongholds = instant win.

(Supply/intel), or a **gang-up** on the leader. GM adjudicates; **betrayal allowed**. Social rubber-band when 3–4 factions.

bridge built **+5**, stronghold captured **+50**. Highest at 90 min wins (or hold-all-strongholds = instant). Live on the LED board.

Time	Phase	Unlocks / events
		visible → gang-up pressure
70–90	Siege	GM rings the bell → Walls batterable; delay towers gate strongholds
GM balance valve : if a faction's lead > threshold, GM posts a bounty — that faction's sectors worth 2× to capture. Transparent, designed rubber-band.		

11. Anti-degenerate design (the levers ↔ the failure modes)

Failure	The rule that prevents it	Failure	The rule that prevents it
Blob (pile on one point)	±5 presence cap + per-sector yield cap	Snowball (leader runs away)	yield cap + decay (must spread thin) + GM bounty + parley gang-up
Turtle (sit and hold)	uncontested decay + territory-over-time scoring	Runner-cheese (one cycler holds map)	control needs <i>distinct present bodies</i> (quorum)
Rush (early stronghold kill)	strongholds invulnerable until Phase 4		

12. Edge cases & rulings

- Net 0 on a sector → meter holds, slow-decays. Hysteresis stops flicker.
 - **Station offline** → marshal falls back to paper headcount; control tent reconciles. (Graceful degradation — the decoupling point.)
- Dead phone / lost band → marshal taps the player in manually; foot soldiers never strictly need a phone.
 - Faction-switching / spectators → badges faction-locked at onboarding.

13. Why this is the sim's job

Every number above is a **lever**: capRate, presence cap, yields, sink costs, Wall, battering rate, phase timings, class weights, bounty threshold. You cannot tune 13

interacting levers on a 150-person field. You tune them in the sandbox (Monte-Carlo + RL self-play), ship a near-balanced rulebook, and spend the real pilots on what the sim can't see: whether civilians have fun.